KnowledgeOS Enterprise Test Data — Technical Architecture & Developer Guide

Field	Value
Version	5.2
Classification	INTERNAL - ENGINEERING & TECHNICAL STAFF
Effective Date	March 1, 2024
Last Reviewed	February 15, 2024
Document Owner	VP Engineering / Chief Architect
Review Cycle	Quarterly
Audience	Software Engineers, Platform Engineers, SREs, Security Engineers, Data Engineers

---

1. Architecture Overview

1.1 System Context

TechGlobal Solutions operates a multi-tenant B2B SaaS platform ("Nebula Platform") serving 2,000+ enterprise customers processing 50B+ API requests/month across 40+ countries. The platform comprises 200+ microservices deployed on Kubernetes across 3 cloud providers with a unified control plane.

1.2 Architectural Principles

- Cloud-Native First: Containers, Kubernetes, GitOps, Observability-driven
- Multi-Cloud by Design: No vendor lock-in; portable workloads (AWS primary)
- Zero Trust: Never trust, always verify; identity-centric security
- Data Gravity: Compute follows data; regional data residency
- Platform Thinking: Self-service infrastructure; golden paths; paved roads

- Evolutionary Architecture: Fitness functions; automated governance; tech debt budget

1.3 High-Level Topology


GLOBAL EDGE (CloudFlare/AWS CloudFront)
DDoS Protection | WAF | Bot Management | Edge Caching | Geo-Routing
API GATEWAY LAYER (Kong/Krakend)
AuthN/AuthZ | Rate Limiting | Request Routing | Protocol Translation
CONTROL PLANE (Global)
DATA PLANE (Regional)
PLATFORM SERVICES (Shared)
- Tenant Mgmt
- Config/Feature Flags
- Deployment Orchestration
- Policy Engine (OPA)
- API Services
- Stream Processing
- ML Inference
- Batch/ETL
- Identity (Okta)
- Secrets (Vault)
- Messaging (Kafka)

```
Observability
    - Audit/Compliance        - Search (OpenSearch)       - CI/CD
(ArgoCD)
```

2. Technology Stack

2.1 Languages & Runtimes (Approved - Golden Path)

Category	Primary	Supported	Deprecated/Restricted
Backend Services	Go 1.22+	Java 21, Rust	Node.js (legacy only)
Data/ML	Python 3.11+	Scala 2.13 (Spark)	R (research only)
Frontend	TypeScript 5.3+	React 18, Next.js	Vue, Angular (legacy)
Infrastructure	Terraform, Go	Python (operators)	Bash (scripts only)
Mobile	Swift, Kotlin	React Native	-

2.2 Frameworks & Libraries (Standardized)

- Go: Gin/Gonic (HTTP), gRPC, sqlc, testify, zap (logging), viper (config)
- Java: Spring Boot 3.2, Micrometer, Resilience4j, Testcontainers
- Python: FastAPI, Pydantic, Polars, PyTorch, LangChain, Great Expectations
- TypeScript: React, TanStack Query, Zod, Playwright, Storybook

2.3 Data Stores (By Use Case)

Use Case	Primary	Alternatives	Notes
Transactional (OLTP)	PostgreSQL 16 (Aurora)	Cloud SQL, AlloyDB	Multi-AZ, Global DB for Tier 0
Document/NoSQL	MongoDB Atlas	DynamoDB, Cosmos DB	TTL, change streams
Cache/Session	Redis Cluster (ElastiCache)	Valkey, Dragonfly	TLS, ACL, cluster mode

Search/Log Analytics	OpenSearch (managed)	Elastic Cloud	ISM policies, cross-cluster
Time Series/Metrics	VictoriaMetrics	Prometheus, InfluxDB	Long-term storage, downsampling
Analytics/OLAP	Snowflake	BigQuery, Redshift	Zero-copy cloning, time travel
Event Streaming	Apache Kafka (Confluent Cloud)	Redpanda, MSK	Schema Registry (Avro/Protobuf)
Graph/Relationships	Neo4j Aura	Neptune	Fraud detection, recommendations
Object Storage	S3 (Intelligent Tiering)	GCS, Azure Blob	Versioning, Object Lock, CRR

2.4 Infrastructure & Platform

- Orchestration: EKS (AWS), GKE (GCP), AKS (Azure) — Cluster API for lifecycle
- Service Mesh: Istio (ambient mode) — mTLS, traffic management, authorization
- GitOps: ArgoCD (App of Apps) — SealedSecrets, Kustomize, Helm
- IaC: Terraform (modules in GitHub, Atlantis for PR automation)
- Policy: OPA Gatekeeper (admission), Kyverno (mutating/validating)
- Secrets: HashiCorp Vault (dynamic secrets, transit encryption, PKI)
- Networking: Cilium (CNI, NetworkPolicy, L7), Envoy (sidecar/waypoint)
- DNS: External-DNS (Route53/CloudDNS), CoreDNS (internal)

2.5 Observability Stack (Unified)

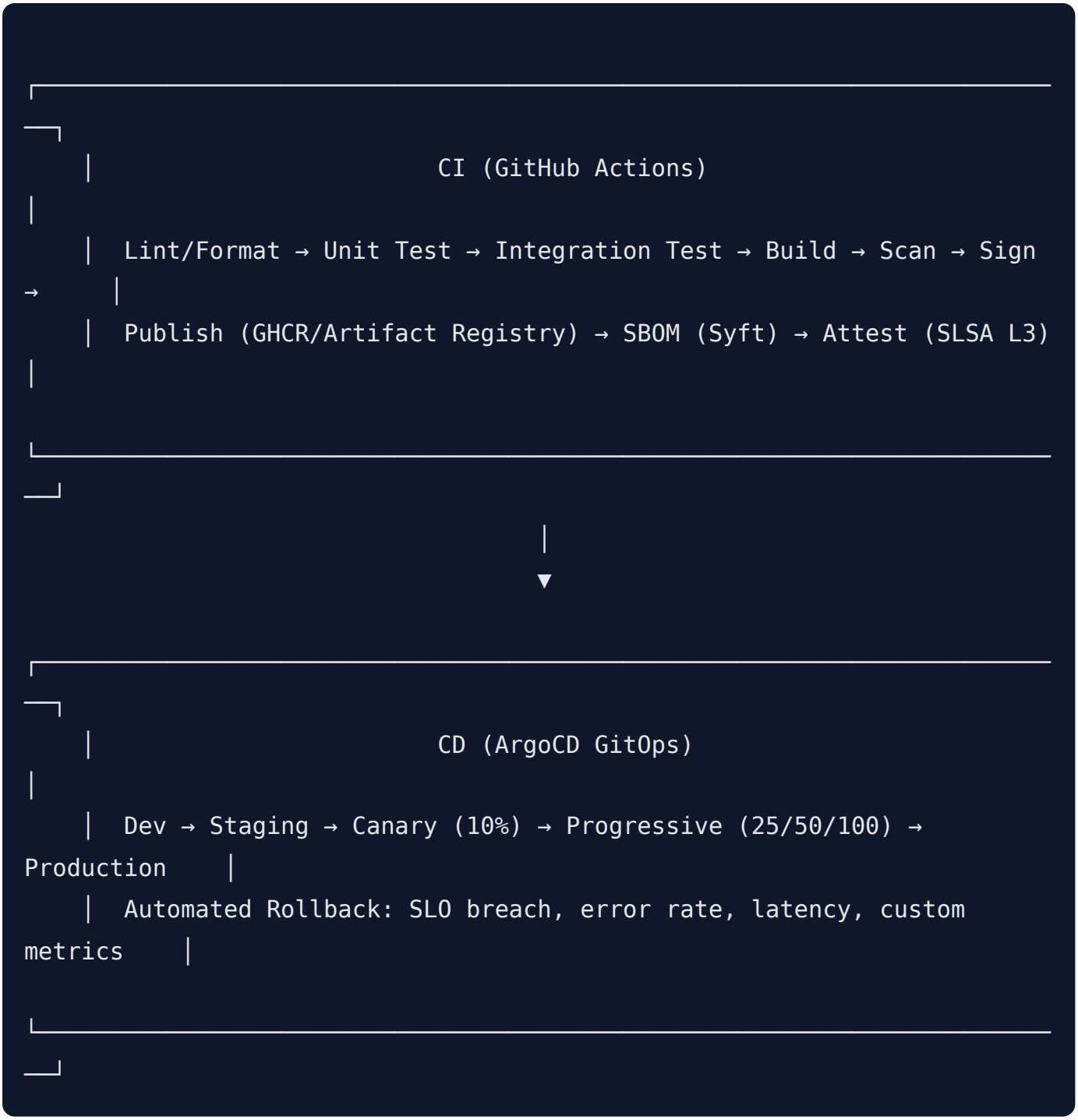
- Metrics: VictoriaMetrics (Prometheus-compatible) + Grafana Cloud
- Logs: Loki (Grafana) + OpenSearch (compliance/analytics) — structured JSON
- Traces: Tempo (Grafana) + OpenTelemetry Collector (OTel) — W3C TraceContext
- Profiling: Pyroscope (continuous profiling)
- Alerting: Alertmanager + Grafana OnCall + PagerDuty
- SLOs: Nobl9 (SLO platform) — error budgets, burn rate alerting
- RUM: Grafana Faro (frontend) + OpenTelemetry (mobile)

3. Development Workflow

3.1 Source Control (GitHub Enterprise)

- Org: github.com/techglobal
- Branching: Trunk-based development (main branch only)
- Short-lived feature branches (< 2 days, < 400 lines)
- Commit Convention: Conventional Commits (feat:, fix:, refactor:, docs:, chore:)
- Signing: Sigstore/cosign keyless signing (OIDC)
- Protection: Required reviews (2), status checks, linear history, CODEOWNERS

3.2 CI/CD Pipeline (GitHub Actions → ArgoCD)



3.3 Environments

Environment	Purpose	Data	Access	Deploy Trigger
Local	Developer inner loop	Synthetic	Full (dev)	Manual (tilt/devspace)
Preview	PR validation	Subset/Anon	PR author + reviewers	Auto on PR open

Dev	Integration testing	Anonymized	Engineering	Auto on merge
Staging	Pre-prod validation	Prod-like	Engineering + QA + PM	Auto on tag (rc)
Canary	Production traffic (10%)	Production	Limited (feature flags)	Manual promotion
Production	Live traffic	Production	On-call + Platform (break-glass)	Manual promotion

3.4 Local Development

- Tooling: DevSpace (Kubernetes sync), Tilt (multi-service), Telepresence (intercept)
- Data: Synthetic data generators (per service), Tonic.ai (anonymized prod snapshots)
- Mocks: WireMock (HTTP), MockK (Kafka), LocalStack (AWS), Testcontainers (integration)
- Debugging: Delve (Go), Debugpy (Python), Chrome DevTools (Node), IntelliJ (Java)

4. Service Design Standards

4.1 Service Template (Cookiecutter)

- Structure: cmd/, internal/, pkg/, api/, deployments/, tests/, docs/
- Config: viper (env, file, Vault), feature flags (LaunchDarkly/Unleash)
- Observability: OTel auto-instrumentation + custom spans/metrics
- Health: /healthz (liveness), /readyz (readiness), /metrics (Prometheus)
- Graceful Shutdown: SIGTERM handling, drain connections, flush buffers
- Versioning: Semantic (vMAJOR.MINOR.PATCH), API version in URL (/v1/, /v2/)

4.2 API Design (REST + gRPC)

- REST: OpenAPI 3.1 (source of truth), JSON:API conventions, cursor pagination
- gRPC: Protobuf (buf for linting/breaking change detection), Connect-RPC for web
- Standards: nouns not verbs, plural collections, consistent error format (RFC 9457)
- Versioning: URL versioning (major), header negotiation (minor), backward compatible
- Documentation: Auto-generated (Redocly), examples in spec, contract tests (Pact)

4.3 Event-Driven Architecture

- Event Format: CloudEvents 1.0 (Avro schema in Schema Registry)
- Patterns: Event Sourcing (core domains), CDC (Debezium), Outbox (transactional)
- Topics: `<domain>.<entity>.<action>` (e.g., billing.invoice.created.v1)
- Guarantees: At-least-once delivery, idempotent consumers, exactly-once via transactions
- Consumer Groups: Per service per environment; dead letter topics with retry/backoff

4.4 Data Modeling & Migrations

- ORM: sqlc (Go), SQLAlchemy/Alembic (Python), JOOQ/Hibernate (Java)
- Migrations: Versioned, reversible, backward compatible, zero-downtime
- Strategy: Expand/Contract (add column → deploy → remove old)
- Testing: Migration tests in CI, staging dry-run, pgRoll for complex changes

5. Security by Design

5.1 Application Security (Shift Left)

- SAST: Semgrep (CI), CodeQL (scheduled), custom rules
- SCA: Dependabot (auto-PR), Syft (SBOM), Grype (vuln scan), Renovate (updates)
- Container: Trivy (CI), Cosign (sign/verify), Kyverno (admission policies)
- Secrets: TruffleHog (pre-commit + CI), GitGuardian (real-time)
- DAST: OWASP ZAP (staging), Nuclei (templates), scheduled
- Pen Test: Annual external + quarterly internal (Cobalt.io/ Bugcrowd)

5.2 Runtime Security

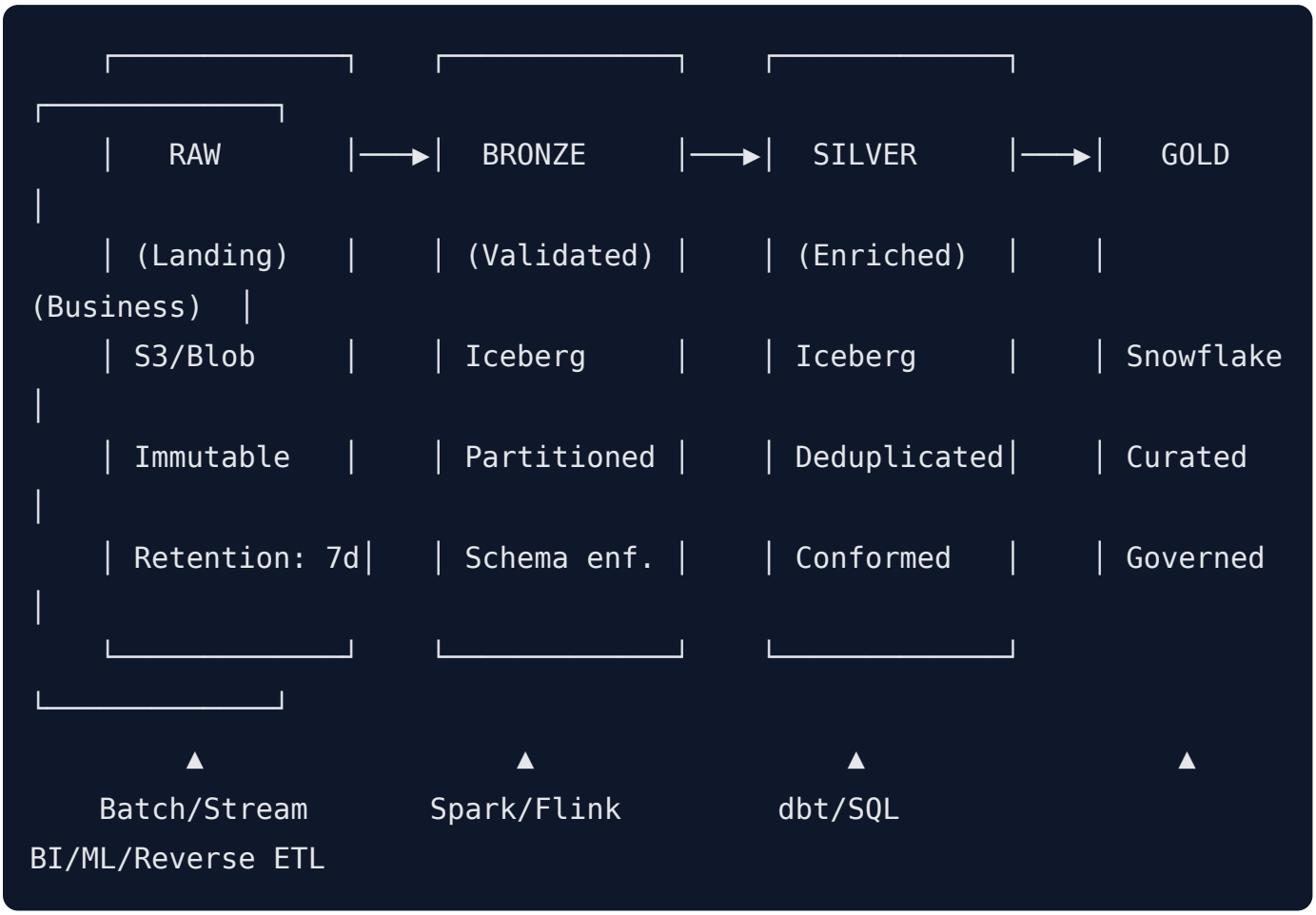
- Admission: Kyverno (no privileged, read-only rootfs, non-root, drop caps)
- Runtime: Falco (syscall), Tracee (eBPF), Cilium (L7 NetworkPolicy)
- Image: Distroless/Chainguard (minimal), SBOM attached, SLSA Level 3
- Supply Chain: Sigstore (keyless), in-toto attestations, Rekor transparency log

5.3 Data Protection (Code Level)

- PII: Field-level encryption (Vault Transit), tokenization, masking in logs
- Classification: Structured logging with data_classification field
- DLP: Custom OTel processor (regex/ML) → redact before export
- Key Rotation: Automated (Vault), 90-day max, zero-downtime rotation

6. Data Engineering & ML Platform

6.1 Data Architecture (Medallion)



6.2 Orchestration & Transformation

- Orchestration: Airflow (managed, MWAA) + Dagster (Python-native, asset-based)
- Transformation: dbt Core (SQL), Spark/Flink (Python/Scala), SQLMesh (emerging)
- Quality: Great Expectations (contracts), Soda Core, dbt tests, Monte Carlo (observability)
- Catalog: DataHub (metadata, lineage, ownership, glossary)
- Contracts: Schema contracts (Protobuf/Avro), SLA contracts (freshness, completeness)

6.3 ML/LLM Platform

- Experimentation: MLflow (tracking, registry, projects) + Weights & Biases
- Feature Store: Feast (online/offline), Tecton (managed) — feature definitions as code
- Training: Ray (distributed), Kubeflow Pipelines, SageMaker (managed)
- Serving: KServe (v2 inference), BentoML, Triton, vLLM/TGI (LLM)
- LLMOps: LangSmith (tracing), Ragas (eval), Guardrails (validation), PromptLayer

- Vector DB: pgvector (PostgreSQL), Pinecone, Weaviate — hybrid search (BM25 + ANN)
- Governance: Model cards, bias reports, drift monitoring, A/B testing (GrowthBook)

7. Reliability Engineering (SRE)

7.1 Service Level Objectives (SLOs)

Service Tier	Availability	Latency (p99)	Error Rate	Throughput
Tier 0	99.99%	< 100ms	< 0.01%	100K RPS
Tier 1	99.95%	< 200ms	< 0.05%	50K RPS
Tier 2	99.9%	< 500ms	< 0.1%	10K RPS
Tier 3	99.5%	< 1s	< 0.5%	1K RPS

- Error Budget: 28-day rolling window; burn rate alerting (2%/1h, 5%/6h, 10%/1d)
- Consequence: Budget exhausted → freeze deployments (except fixes), focus on reliability

7.2 Incident Response (Integration with Security IRP)

- On-Call: Follow-the-sun (3 regions), 1 primary + 1 secondary per service
- Escalation: PagerDuty (Tier 0: 2min, Tier 1: 5min, Tier 2: 15min)
- Runbooks: Executable (RunWhen/RunMe), versioned with service, tested quarterly
- Blameless Postmortems: Template, timeline, RCA, action items (Jira), shared widely

7.3 Chaos Engineering

- Tooling: Chaos Mesh (K8s), Gremlin (infrastructure), LitmusChaos
- Experiments: Pod kill, network latency/loss, CPU/memory pressure, clock drift, DNS failure
- Cadence: Weekly (automated), Monthly (game day), Quarterly (cross-team)
- Safety: Blast radius controls, auto-abort on SLO breach, production only with approval

7.4 Capacity Planning

- Forecasting: Prophet (time series), 90-day horizon, 95th percentile
- Autoscaling: KEDA (event-driven), VPA (vertical), Cluster Autoscaler (nodes)
- Load Testing: k6 (CI), Gatling (pre-release), Locust (continuous)
- Rightsizing: Goldilocks (VPA recommender), Kubecost (cost allocation), quarterly review

8. Developer Experience (DevEx)

8.1 Platform Services (Self-Service)

- Service Scaffold: `platform-cli new service` → PR with CI/CD, docs, monitoring
- Database: `platform-cli db provision` → PostgreSQL/Mongo/Redis (namespaced, templated)
- Messaging: `platform-cli topic create` → Kafka topic (partitions, retention, ACLs)
- Secrets: `platform-cli secret sync` → Vault → K8s (External Secrets Operator)
- Feature Flags: LaunchDarkly SDK + Platform wrapper (typed, audited)
- Deploy: `platform-cli deploy <env>` → ArgoCD application sync

8.2 Inner Loop Optimization

- Build: Earthly (reproducible, cached), Nix (dev environments), Devbox
- Test: Unit (< 10s), Integration (< 60s), Contract (< 30s) — parallelized
- Feedback: PR checks < 5 min (target), flaky test detection/quarantine
- Docs: Docs-as-code (MkDocs/Material), ADRs (MADR format), API docs auto-published

8.3 Productivity Metrics (SPACE Framework)

- Satisfaction: Quarterly DevEx survey (eNPS, tooling, process)
- Performance: Lead time (PR→prod), deploy frequency, change failure rate, MTTR (DORA)
- Activity: PRs, commits, reviews, build/test times
- Communication: Collaboration index, knowledge sharing, onboarding time
- Efficiency: Focus time, meeting load, context switching, toil reduction

9. Governance & Compliance (Automated)

9.1 Policy as Code

- OPA/Rego: Kubernetes admission, Terraform plan, CI gates, runtime (Gatekeeper)
- Examples: No public S3 buckets, encrypted volumes, resource limits, approved images
- Drift Detection: Cloud Custodian (cloud), Kyverno (K8s), Terraform drift (Atlantis)
- Remediation: Auto-remediate (low risk), ticket + notify (medium), page (high)

9.2 Compliance Automation

- SOC 2: Continuous control monitoring (Vanta/Drata) — evidence collection automated

- PCI DSS: Network segmentation validation, ASV scans, SAQ automation
- GDPR: Data flow mapping (DataHub), DPIA triggers, DSR automation, RoPA updates
- FedRAMP: NIST 800-53 control inheritance, POA&M tracking, continuous monitoring

9.3 Architecture Review Board (ARB)

- Charter: New services, major changes, cross-cutting concerns, tech debt > $50K
- Cadence: Bi-weekly; async RFC process (GitHub Discussions) for < $10K
- Template: Context, Decision, Consequences, Alternatives, Risks, Migration Plan
- Decisions: ADRs (Architecture Decision Records) — immutable, versioned, linked

10. Cost Management (FinOps)

10.1 Cost Visibility

- Allocation: Kubecost (K8s), CloudHealth (multi-cloud), Snowflake (data)
- Chargeback: Per service, per team, per environment, per customer (Tier 0)
- Anomalies: Daily ML-based detection (CloudForecast), Slack alerts
- Budgeting: Quarterly planning, monthly variance analysis, rolling forecast

10.2 Optimization Strategies

- Compute: Spot/Preemptible (batch, dev), Savings Plans (steady), Right-sizing
- Storage: Tiering (S3 Intelligent Tiering), lifecycle policies, compression
- Data: Partition pruning, clustering, materialized views, query optimization
- Network: VPC endpoints, CloudFront, regional deployment, data transfer optimization
- K8s: Bin packing, node pools (ARM/Graviton), sleep mode (dev), cluster sharing

10.3 Unit Economics

- Metrics: Cost per API request, per active tenant, per GB processed, per MAU
- Targets: < $0.0001/request (Tier 0)$, <0.10/MAU, < $50/TB processed
- Reviews: Monthly (FinOps), Quarterly (Exec), Continuous (engineering ownership)

11. On-Call & Operations

11.1 On-Call Structure

- Rotation: Weekly (Mon-Mon), follow-the-sun (US/EU/APAC)
- Compensation: $500/week (primary)$,250/week (secondary), $100/incident (page)

- Handoff: Structured (runbook review, open incidents, deployments, risks)
- Override: PagerDuty (vacation, illness), minimum 24hr notice

11.2 Runbook Standards

- Format: Markdown + executable (RunWhen), versioned in service repo
- Sections: Symptoms, Diagnosis, Resolution, Verification, Escalation, Links
- Testing: Quarterly execution (game day), post-incident updates
- Ownership: Service team owns; Platform team owns platform runbooks

11.3 Change Management

- Standard Changes: Pre-approved (dependency updates, config, scaling) — auto-merge
- Normal Changes: PR + approval (1 reviewer) + scheduled window (Tue-Thu 10-16 UTC)
- Emergency Changes: Break-glass (CISO/VP Eng), post-implementation review (48 hrs)
- Freeze: Code freeze (Dec 20-Jan 5), pre-earnings (2 weeks), major events

12. Reference Architectures (Patterns)

12.1 New Microservice Checklist

- [ ] Scaffold from template (Cookiecutter)
- [ ] OpenAPI spec + Protobuf definitions
- [ ] Unit tests (> 80% coverage), Integration tests (Testcontainers)
- [ ] CI/CD pipeline (GitHub Actions → ArgoCD)
- [ ] Observability: OTel, metrics (RED), logs (structured), traces
- [ ] Health endpoints, graceful shutdown, config (Vault/Env)
- [ ] Feature flags, secrets, database migration strategy
- [ ] Security: SAST/SCA/Container scan, threat model (STRIDE)
- [ ] Documentation: README, ADR, API docs, runbook
- [ ] Load test, chaos experiment, cost estimate
- [ ] ARB review (if required), Platform team sign-off

12.2 Async Processing Pattern

```
API → Validate → Persist Outbox (DB) → CDC (Debezium) → Kafka → Consumer
→ Process
```

- Guarantees: At-least-once, idempotency keys, exactly-once via transactions
- Monitoring: Lag alerts, processing latency, dead letter queue depth

12.3 Multi-Tenant Data Isolation

- Row-Level Security (PostgreSQL RLS) + Tenant Context (middleware)
- Schema-per-tenant (legacy) → Shared schema with RLS (standard)
- Encryption: Per-tenant keys (Vault), envelope encryption for large objects
- Noisy Neighbor: Resource quotas (K8s), query limits, priority classes

13. Key Repositories & Resources

Repository/Resource	Purpose	Owner Team
platform/cookiecutter-service	Service template	Platform
platform/terraform-modules	IaC modules (VPC, EKS, RDS, etc.)	Platform
platform/helm-charts	Helm charts (services, infrastructure)	Platform
platform/argo-apps	ArgoCD Applications (App of Apps)	Platform
platform/policies	OPA/Kyverno policies	Security + Platform
platform/cli	Developer CLI (platform-cli)	Platform
architecture/adrs	Architecture Decision Records	Architecture
architecture/rfcs	Request for Comments (proposals)	Architecture
docs/tech-specs	Technical specifications	Respective Teams
runbooks/	Operational runbooks (per service)	Service Teams
data/contracts	Data contracts (schemas, SLAs)	Data Engineering
ml/models	ML model registry (MLflow)	ML Platform

14. Escalation & Support

14.1 Engineering Support Tiers

- L1: Platform Team (Slack #platform-support, 9-5 business hours)

- L2: Platform On-Call (PagerDuty, 24/7 for platform infra)
- L3: Architecture Review Board (RFCs, major decisions)
- Security: #security-help (Slack), security@ (email), IRP for incidents

14.2 Getting Help

- Documentation First: docs.techglobal.internal (Confluence + GitHub Pages)
- Slack: #platform-support, #devops, #security-help, #data-help, #ml-help
- Office Hours: Platform (Tue/Thu 2pm), Security (Wed 11am), Data (Mon 3pm)
- Ticket: ServiceNow (platform requests), Jira (bugs/features)

15. Glossary

- **ADR:** Architecture Decision Record
- **CDK:** Cloud Development Kit
- **CDC:** Change Data Capture
- **CNI:** Container Network Interface
- **CRD:** Custom Resource Definition
- **CSI:** Container Storage Interface
- **CSPM:** Cloud Security Posture Management
- **CWPP:** Cloud Workload Protection Platform
- **DORA:** DevOps Research and Assessment (metrics)
- **eBPF:** Extended Berkeley Packet Filter
- **FinOps:** Financial Operations
- **GitOps:** Git as source of truth for declarative infrastructure
- **HPA:** Horizontal Pod Autoscaler
- **IaC:** Infrastructure as Code
- **IRP:** Incident Response Plan
- **KEDA:** Kubernetes Event-driven Autoscaling
- **MTTR:** Mean Time To Recovery
- **OPA:** Open Policy Agent
- **OTel:** OpenTelemetry
- **PII:** Personally Identifiable Information
- **RFC:** Request for Comments
- **RPO:** Recovery Point Objective

- **RTO:** Recovery Time Objective
- **SBOM:** Software Bill of Materials
- **SLA:** Service Level Agreement
- **SLO:** Service Level Objective
- **SLSA:** Supply Chain Levels for Software Artifacts
- **SRE:** Site Reliability Engineering
- **VPA:** Vertical Pod Autoscaler
- **WAF:** Web Application Firewall

16. Revision History

Version	Date	Author	Description
5.2	2024-03-01	VP Eng + Architect	Multi-cloud, ambient mesh, SLSA L3, DataHub, ML platform, FinOps, DevEx
5.1	2023-09-15	Platform + Arch	Istio ambient, ArgoCD App of Apps, OPA Gatekeeper, VictoriaMetrics, Pyroscope
5.0	2023-03-01	VP Eng	Major rewrite: cloud-native, GitOps, service mesh, observability unification

---

END OF DOCUMENT